SAMPLE



Dear family,

This document contains a private key that can unlock files encrypted with the associated public key. It is intended as a backup so **must** be kept in at least one secure place, e.g. a safe-deposit box, fire safe, a trusted friend or family member. Instructions for using the key to unlock a file are provided. The instructions are a bit technical, so it includes some ways to get help if needed.

Know that the key *itself* is encrypted, so it **cannot** be used without the passphrase used to create it. The passphrase may be written on the last page. If it is not, the Notes on that page say where to find it. If you don't have or can't get the passphrase, so sorry but there is nothing you can do but stop here.

Welcome to the modern world of digital security.

If all of this has you thinking *"uh... what?"*, please see Getting Help.

I hope you never need this!

Alex Sample

Getting Help

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The following are contacts that I trust and who have agreed to help if needed:

1. **Jordan Lee**

   Phone +1 555 010 4477
   Email jordan.lee@example.com *(preferred)*
   Address 12 Elm Street, Springfield, OR 97477, USA

2. **Sam Rivera**

   Email sam@example.org
   Social Signal @samrivera

If they aren't available, don't worry – there are people out there who can help you through this process. Since what you're trying to decrypt may be extremely sensitive (e.g. a password manager backup), you need to be careful about who you trust and how you get help...

Finding Help

Your Best Options: In-Person Help

Friends or Family in Tech: Think about anyone in your network who works with computers professionally – software developers, IT support, network administrators, or even tech-savvy relatives. They might be willing to help, and you already know them personally.

Local Computer Repair Shops: This is probably your best bet. Most towns have at least one computer repair shop, and many of their technicians know their way around the command line. When you call, you can say something like: "I need help following some written instructions to run commands in a terminal. It involves decrypting a very sensitive file, so I need someone who can work on this in person and agrees not to keep any copies of my files or information."

Computer Store Support: Many computer stores and manufacturers offer in-store help (for example, the Genius Bar at an Apple Store). While they might not help with the actual decryption, they can help you get comfortable with opening a terminal and understanding basic navigation. Make an appointment and explain that you need to learn some basic command line skills for accessing your own files.

Local College IT Departments: If you have a community college or university nearby, their IT support staff are usually knowledgeable and trustworthy. You might be able to arrange a consultation where they walk you through the process.

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Community Resources

Library Technology Centers: Many public libraries have tech help sessions or staff who assist with computer problems. While they might not know the specific commands, they can help you get comfortable with the basics.

Senior Centers or Community Centers: If you're looking for a comfortable, non-intimidating environment, these often have volunteer tech helpers who are patient and understanding.

Last Resort: Online Help

Only if you absolutely cannot find in-person help, you might consider freelance platforms like Upwork or Fiverr. But please understand – this is risky because what you're decrypting may be extremely sensitive, such as a password manager backup or financial records. If you go this route:

- Never, ever share your actual files or private keys
- Only ask for general guidance on the process
- Work with someone who has excellent reviews and a long track record
- Consider having them walk you through the steps over video call rather than doing the work themselves

What to Say to Anyone Helping You

Here's some language you can use to protect yourself:

Friends and family

"I need your help unlocking a really sensitive file. Before we get started, I just want to ask that you don't make any copies, take screenshots, or save anything from this session. It has deeply personal information, and I'm trusting you to keep it completely private. I really appreciate your help."

Professionals

"I need help with a very sensitive file decryption. Before we start, I need your agreement that you will not make any copies of my files, take screenshots, or retain any information from this session. The file contains extremely sensitive personal data, and I need your word that everything stays confidential and nothing gets saved to your system."

Professionals should understand completely and agree without hesitation.

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Red Flags – When to Walk Away

Trust your instincts. If someone:

- Asks to take your computer home or work on it remotely
- Wants to copy your files “for backup purposes”
- Seems evasive about confidentiality agreements
- Pressures you to rush through the process
- Asks too many questions about what’s in the encrypted file

Just politely decline and find someone else.

A Final Thought

Remember, you’re not asking for anything unreasonable. Plenty of people know how to use the command line, and most will be happy to help you follow written instructions. The key is finding someone trustworthy who respects the sensitive nature of what you’re working with. Take your time, and don’t settle for help that doesn’t feel right.

Instructions

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This section assumes familiarity with the terminal and using command line tools. The tools described here run on macOS and Linux, and on Windows in its Linux environment, WSL (see Installing Nix).

If *you* are helping

These are Alex Sample's files, and they may be extremely sensitive. Anyone holding this document can try to decrypt them, so before you start, make sure that the person asking should have access:

- They are someone Alex would want to have it, such as family or the executor of their estate.
- They need it now. If Alex can be reached, check with them first, using contact details you already have rather than any the person asking gives you.
- Their story holds up. If they are a stranger to you, or anything seems wrong, stop; pressure to hurry is a warning sign in itself.

Prerequisites

The private key and the passphrase used to encrypt it are required. Hopefully you have the key in digital form, perhaps on a USB/thumb drive or in a file (it should have a '.key' file suffix). If you don't, this document's Paper Key pages hold two forms of backup: a QR code and ("`PEM`" formatted) text, both containing the encrypted private key. See Getting the Key File for how to make the key file from either.

The passphrase used when the key was created is either written on the Passphrase page of this document, or kept separately, in which case the Notes section says where to find it.

You need **one** of the following:

- Nix (recommended), which downloads and runs the valt recovery tool for you, with one command. It always uses the versions of age and the other tools that valt has tested with (not whatever the computer happens to have), installs nothing else, and can be removed afterwards. See Installing Nix, below.
- `valt`, the tool used to create this document. Source and instructions can be found at github.com/phoggy/valt.
- A command line tool that performs age file encryption and decryption (the format is described at c2sp.org/age), such as age 1.3.0 or later, from github.com/FiloSottile/age, plus `tar`, which is included with macOS and Linux, and optionally `minisign`, to check signatures, from github.com/jedisct1/minisign. Another age implementation will work, so long as it supports post-quantum (`-pq`) keys.

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Identifying Your File

The end of the file name tells you what kind of file you have. File browsers often hide it; running `ls` in a terminal shows full names.

- **`.age`**: a single encrypted file. A file with the same name plus `.minisig` may be next to it; keep the two in the same directory so the signature can be checked.
- **`.valt`**: a secure archive, which can contain any number of files and directories. A file with the same name plus `.pub` may be next to it; it contains no secrets and can be ignored.

Getting the Key File

The commands below need the private key as a file: `KEY_FILE` in them. If you have it already (a file ending in `.key`, perhaps on a USB drive), use that. Otherwise, make it from this document's Paper Key pages near the end, which hold the same encrypted key as a QR code and as text, in one of these three ways:

Decode QR Code

The QR code contains binary values so the usual phone camera decoding will likely not work. Instead, take a photo and move it to your computer. Use `qrtool` to convert the image (PNG, JPEG, etc.) to a key file:

1. `nix run github:phoggy/valt#qrtool -- decode IMAGE_FILE > KEY_FILE` or
2. `brew install qrtool; qrtool decode IMAGE_FILE > KEY_FILE`

The result should match the checksum shown in the next method.

Scan PEM Text

Scan the PEM text and send/save it as a file. If you don't know how, search the web for "scanning text on phone". Verify that the result matches the checksum to guard against scanning errors:

```
sha256sum KEY_FILE | head -c 16
```

It should match

```
1a8ca629d4c1116d
```

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Manual PEM Text Entry

If neither of the first two options works, you can always type it in by hand. Yes, this is *extremely* tedious, but it *can* be done. Be sure to include the leading and trailing lines, each of which begins and ends with 5 dashes. Use the checksum test from the previous method to ensure it was entered correctly!

Installing Nix

Nix fetches software and runs it, without changing anything else on the computer. Install it with the installer for your computer:

- **Mac with Apple silicon** (Apple menu, About This Mac, shows a chip such as Apple M1): download and run the installer from dtr.mn/determinate-nix.
- **Linux**: in a terminal, run:

  ```
  curl --proto '=https' --tlsv1.2 -sSf -L https://install.determinate.systems/nix |
  sh -s -- install
  ```

- **Intel Mac**: in a terminal, run:

  ```
  curl -L https://nixos.org/nix/install | sh
  ```

  With this installer, each `nix` command in this document needs this option right after `nix`:

  ```
  --extra-experimental-features 'nix-command flakes'
  ```

- **Windows**: Nix runs in Windows' own Linux environment, WSL. Open PowerShell as administrator, run `wsl --install`, restart, open Ubuntu from the Start menu, and follow the Linux step above inside it (see learn.microsoft.com/windows/wsl/install). Your Windows files are under `/mnt/c` there, e.g. `/mnt/c/Users/NAME/Downloads`.

Installing asks for an administrator password and takes a few minutes; on a Mac it creates a separate volume for Nix. Answer yes to its questions and allow any system dialogs that appear. When it is done, **open a new terminal window**: one that was already open does not know about Nix yet.

The first time the recovery tool runs, it downloads what it needs, about 1 GB, so it needs an internet connection and may take several minutes. Later runs start at once.

To remove Nix afterwards, run `/nix/nix-installer uninstall` if you used the first two installers; after the Intel Mac one, follow nix.dev/manual/nix/latest/installation/uninstall.

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Decrypting With The Recovery Tool

With Nix installed (see Installing Nix, above), open a terminal, change to the directory containing the encrypted file or archive, and run:

```
nix run github:phoggy/valt#recover -- decrypt FILE -i KEY_FILE
```

Replace `FILE` with the `.age` or `.valt` file and `KEY_FILE` with the private key file (see Getting the Key File). You will be prompted for the passphrase that protects the private key.

- An encrypted file is decrypted next to the original, without the `.age` suffix. If a signature file is present it is checked first, and decryption stops if the file has been tampered with.
- A secure archive is verified and its contents are extracted into the current directory. Use `-o DIRECTORY` to extract somewhere else.

This downloads and runs the recovery tool automatically; no other software needs to be installed. If you used the Intel Mac installer (or Nix reports that flakes are not enabled), add this right after `nix`:

```
--extra-experimental-features 'nix-command flakes'
```

If that fails, which a later change to valt could cause, try the valt release current when this document was made:

```
nix run github:phoggy/valt/v0.1.1#recover -- decrypt FILE -i KEY_FILE
```

An archive's `README.txt` gives the same command for the release current when it was made.

Decrypting Using valt

If you installed `valt`, the equivalent is:

```
valt decrypt FILE -i KEY_FILE
```

`FILE` and `KEY_FILE` are as above, and so are the results.

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Decrypting Manually

For an encrypted file, using `age`, and `minisign` to check its signature:

1. If a signature file (`FILE.age.minisig`) is next to it, check that the file has not been tampered with. The public key that checks the signature is inside the private key file; this extracts it (asking for the passphrase), then checks the signature:

   ```
   age -d KEY_FILE | sed -n 's/^# \[minisign\.pub\] //p' > minisign.pub
   minisign -V -p minisign.pub -m FILE.age
   ```

   You should see `Signature and comment signature verified`. If not, stop: the file has been changed.
2. Decrypt it:

   ```
   age -d -i KEY_FILE -o DECRYPTED_FILE FILE.age
   ```

For a secure archive, first unpack it, which needs no key:

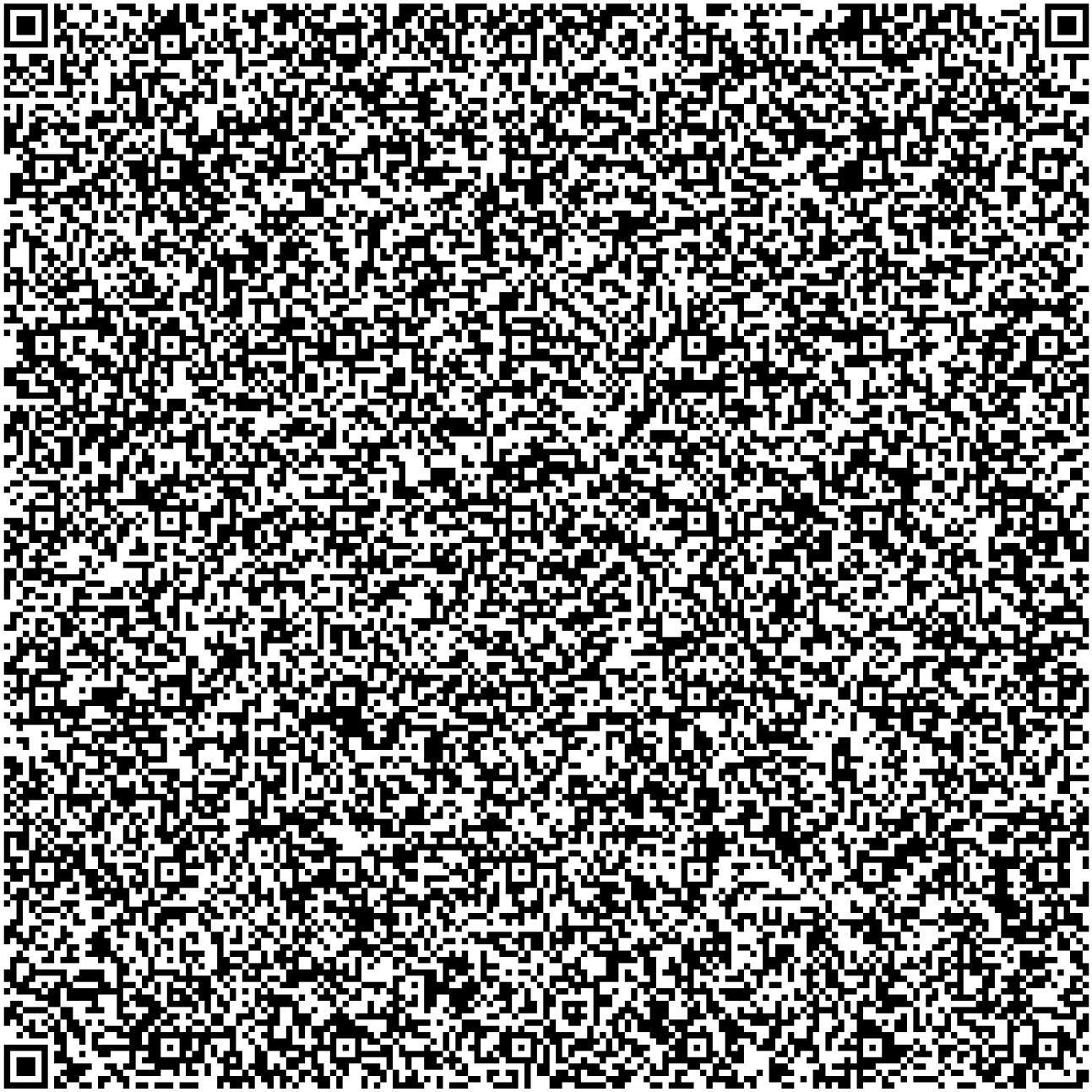
```
tar xJf FILE.valt
```

Then follow the instructions in the `README.txt` file it contains: they use `age` to decrypt, `tar` to unpack, and optionally `minisign` to confirm nothing has been tampered with.

In all cases you will be prompted to enter the passphrase that protects the private key.

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Paper Key



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```
-----BEGIN AGE ENCRYPTED FILE-----
YWdlLWVuY3J5cHRpb24ub3JnL3YxCi0+IHNjcnlwdCAxWTdzMzhIV0RVUFlxektv
M09POWt3IDE4CkxBYTNkbXFQM2p0SUV2b0swY3gyK3lrUmlrMmh3WWZJaW5GSVBv
K2NueFEKLS0tIFdsajl0NW5lV0JPbWVUTUZkK0Zhb09ZdlExb0FkWlBSMFYxQXN5
QVluZEEKlNj7OczZHphLJAqyzUFwdS2t5ZOg8awyuiPSQ6bnL7TEx1LVNkVdngFY
qdJiAwrM3iXWaxnyu3KuUJGno6Le9pcJ8FjAWl2q7fT78IS9kWGA1C2/TmXchP6h
piyC7ZNVun2hQMdODcJLgSZ9Lg4gYbk3zOTuvtlioNjecAEDX88MbpquTKuL2HvC
6bWaWlGiim1c9UUAErNfaNIl2XHUPh/SveaMKdySbiqOVbX6uUDLoSYVL89fFcFx
B4Qg78vFFMAxIUyrsFYrUxYB3PoIBBdjMeu+nrHv+suRdeTnChGBdZpv5Xx0wMUq
3ptoWcPgKdqwsHDl7Qjg/736DaXhWbaFpeLXHs2C5i7IPgG2S3MJQmoDSKXeH6Cw
sOXjEXd/9g7+2dJNZHRVxVzWuLwNxGNfOxpzr621aErqinz9+Z0Mx/nl70pd2JQO
1IBnJWEjszwhZeXSjvl1bI7231Jg4SrUW7V2V268rhLIr7vJEGKDdpsuTM5anQFv
BxQs3g9VDTSu9b7X/d1rzjfHc7Oo54MMGSYIsa6L6BCZ+Ncr22+ckrxkuNotcBW2
c6+i6n32NlsiXJExnqdBmx7O6c5E3JpnZdWKtu3nSdvzGZEfjiOfD+0bZ/oLjmS8
RhHsKCkmYzg3VFVcNKP9MU9Ha8E0EEs+pxSBYsEK7wlVdaMMOtlBQ2cjQowNgGdq
bT+3NEXE4p/3hWTFnabKhOzHZQqYmZmjFpClIJZgYdEQZ9F0X5cJD+bcWmrHwR/w
eu/eDorLpBXFKqvIlb8JjXppDNs9YVWFD5EmXKycau5jo0G7itZpl2btP/W6oPD9
dPTous59f4jbTcvcjmCaDR9zChAJ4UScZydKtSuluEeIPCAxhkdiyJs1XNCpM4CE
ICL5sCXFrKZoZcDLwBMSxU2A0O8gyoWlKpR14BSZZa80yK+yIKB/+0t0+RckdANA
taNSkYbKKLJYXNLdI8g4bQjsIj6GY+lRJJMAi1R2ZprCTSVZ18nSnT7XRKH6GYji
5LN4orHw7bN5xHLVxpu5UrAF8D2Bu+PYw72fjkjCoLEPYy8C/Wak5a3C5gA+D+Wm
4VEv+XAZzheM1XTWzulafR4UxMN3Uqj3w6y32cIqZpcUXDUcYtuFL9knEP47Z+3u
mCUnaQl66Vp+AoVR8uzA2pCLmv16pPXW1lDy7ATo7yiqX2DLk0+C9ltk6ydOrQzy
SCbkdJDb87+xSI9cC/2xkwUXHSJYBDjuBKw1UpGXw7MQlGZOW2rmR5hOf36ckFmP
0WGfhmMqnG7shcP1L/orm3NX6C3nFxKAodoLA+31Qebdn2iWGZZb/7h6yVR5B6si
CNZWtf4h4Olg5HFC2ht4LjSsFVVaMeI8/yALAHn9fr0QSJ9yLhDYLGPgINwU3z/5
4PjAFMhjAcvWb07sVU1eqH5awjZncYo4CkwHkvqipknpV1IACa7Iqk3+Y7vtpbXZ
yZystt1Fzc8++f9lk+4BCqsWmCqRPknqeFyUq4vjhc7DNAQwmuyFTX1UvSN+C+Qb
1jN5H41jakJk1sRz0pyHFnF8u2qODPTzWMle4MLb8f86/SknZBW+VIkqQQSEb92Z
pKDVrfM78RmUPIRmLNf6iwdtmLuafY0LZTYSkMSg/tmUgkflh+BiTztqq7E5POkl
DKzfQo3OhWnDQNhTl79+YdH6Zp4wkvoqRO3E0ZcWQg5chIEVRspkZN3IUa2jglt+
xUo4TrNzoCZGs/V/BHnL8JRvOvWHJ14l6JBDUlxXlnrau1uVKZteZ7I40IeDXjhj
sYo+mGkOB/OSHrsko+TBlPBedQqjNDS9Ydnmdrcjq6Oct7aONA121ccsEDgLQGD3
RL7DBwIXqRWyOw4ASISFGIjTePpXwJCsq1/38XF/dKhRz6aFAq/IMLQJaVBX9wOo
Sr4mlCYBTChDVmWRdoZaHNAaDBDRRcT5lxGKJRLOzFKDdh881yQ8NpoB0z1nqOZ5
WQTxethJxwxGA8uL3SzxakxSAoBB/qYT/7fqKWf79mm9sdu4lnS9DyPJcFmypdJW
RXx8uzVbBBnyhVX6HfLahicly5oY2hMZ4mANopgtCpE155+qSKYbF3P0KbPudN/g
dFAlRFqRVxGMMa0yxlcYJl4xYL+iV4YDPWAPIvVg1MgYmV9YDLXetWWCd5B1RTed
Gvnz44H2mA7gRdD4EoCfntoq5vzRxZyhgmBWEq5X9ammcHPyh2rnsyrsHd6Ewksb
V7Wcy5dfn5fNr7gUowuKuftGwGyGDbowzVZOVEwTkXHwgMySRBduA85VbNwfFIvA
u/buq2RWnfx/sIpPohv/lXVnJ54qH8qWBdf5x0313E6jHbmMbvA2/Z3ermI6NU1L
titPhxHm5fUhL8YEeBSMGZ3PF8d54uOO4mVyTSsxGa2TeB5t6eBcsFeSlG/qK8hr
IIGxNfo9jRX33Wp33g1QeiIC0xaw3KfSeOax2MWTYIMVl9T9c250NhC5Ium0V8Yf
ZlJ7izkT/0ANFkCAWHmBJ7lL3UAg3ZSZeftMDo3awCjg5mkdGL60TJ/DFBkeWVij
qQm9LcfMF7l0Dp0tGI8+A7pLc3hkuaW3X7pBkxohuSg2QxrTIEYIeRqzLeiOm/iR
Okx2ctGjMOZWlx7Vz6CZ8A7THK9dM9tEmS8j/zLHT/3WOoAs6WJ3NrtCx3r2ztkn
3NSToLRyUiIJynHcwdIsLirvWaWfy+kNZByGOZ06AiYXRoKKh8fW1j1wjwhS2iYO
cmawWoIhS0Qlimlvp6FF5BdpIWtkrLRVx9NzNziZxg7JC9c69hFA3lBSGjLS9Nyo
5wyc+LHRbmyF1VuaD96P8YQewRd1aVD/GlVpt6T+55n0s89OkqJseT0+5G/KUKlq
9CAZrELR+PB2Yk6N6caCQQ8rf1zqQzDD1Tme3eHFePhUZICLJ+/NwPn2/B2YjD6o
C2DcTq3RdLjg/pl/q+F03wzMbRE/U4Fmf+/heq51LcGTJSGYSxIuuxUf+1vWjzaI
JomY767iLl8fHOdso87oJY1mThDE5dSrVkX0/tpaCUNt/TjMMvyzL2MkxwtlNyTx
iI1tkhOI3GxoGeKNsuFgf16/YeB7sK+Ci/hDpClsWHaJLKFFRhpBz/yYwyhjsQZB
5Zt5CJuRqrZVIGUkqi6Cfrtrbi8IeVW8lcSS7qVWD5IGS6Np/tF8egXeMYqWO+i9
Mc1Tl32m8EjvpZzctfnM8KdUMCXPoCKKY+wSmrrl53oU4zVIrLyf3/ZRbDtWR5qe
Rhz6nVCjU71V/62pY8069yXIgaVCSMQRRjV9VvN+h4AoW7okrH8uDGVk68HKd4ri
70F+3/B1tGqZgDL+7rpJbnybRRQTpsNy11ok3Nk4/sdr72wdlWxwwoeiVuLbg4wR
38SyFgTuU+tihRoUrndHXuF33ODJU2xBSsRIP5VVjEc=
-----END AGE ENCRYPTED FILE-----
```

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Passphrase

Write the passphrase in the boxes below, one character per box:

Notes

The passphrase is in a sealed envelope with my will, held by my attorney, Pat Morgan. The encrypted backups are on the USB drive labeled 'valt' in the fire safe.